

Nithish K Gnani

📍 Bengaluru, Karnataka, India
🌐 [linkedin.com/in/nithish-k-gnani](https://www.linkedin.com/in/nithish-k-gnani)

✉ nithishkgnani@gmail.com
🐙 GitHub

🌐 Website

EDUCATION

National Institute of Technology Karnataka
Bachelor of Technology - Mechanical Engineering

Surathkal, Karnataka, India
July 2013 - May 2017

SKILLS

- **Programming Languages:** P4, Python, MATLAB, C, Embedded C
- **Hardware :** Netronome Agilio Smart NIC, Nordic nRF52 series, Arduino, NetFPGA-SUME
- **Others :** P5.JS, Adobe Premiere Pro, Autodesk Fusion 360, 3D printing, Implementing papers

EXPERIENCE AND PROJECTS

Indian Institute of Science (IISc)

Bengaluru

Research Associate

March 2019 - Present

- **Time Sensitive Networking (TSN) and Tactile Cyber-Physical Systems:** (Funded by Ministry of Electronics and Information Technology (MeitY) and Cisco - Centre for Networked Intelligence (CNI), IISc)
Design and implementation of a Tactile Cyber-Physical System (TCPS) for real-time interaction between humans and robots for applications requiring ultra-reliable low latency communication (uRLLC). To develop IEEE 802.1 TSN enabled ethernet switches using programmable network hardware like the Netronome Agilio Smart NIC with P4 programming language. Setting up a CPS testbed with a robotic arm and haptic device for demonstrating a tele-operation scenario.
- **Wi-Fi Modelling:** (Client: Boeing) Modelling & simulation of IEEE 802.11n & 802.11ac and rate adaptation algorithms using MATLAB. Performance Metrics: throughput, packet error rate & range. Implementing Rate Control Algorithms in MATLAB.
- **Airplane IoT data analytics and management:** (Client: Boeing) Developed machine learning algorithms to manage the real time data generated from sensors from a Simulink model of an aircraft ECS (Environmental Control System) by adaptive storage of critical data while maintaining excellent data reconstruction capability using Python, under storage-space constrained scenario.
- **Acoustics Based Localization:** (Client: Boeing) Developed an application to locate multiple wireless edge devices with embedded microphones by using audio signals from speakers. *Novelty:* Unlike the usual source localization, here the receivers (mics) are localized when the locations of speakers are known.
- **AMMAZING:** An end to end 5G mmWave system for infotainment. *Novelty:* Data & control plane split over 5G and regular Wi-Fi. Currently shortlisted in top 10 teams in ongoing 5G Hackathon by DoT, GoI.

Hindustan Coca-Cola Beverages Private Limited

Hyderabad

Team Leader, Maintenance

July 2017 - April 2018

- **Total Productive Maintenance (TPM):** Implemented TPM steps 4 and 5 in the Ameenpur bottling plant which led to the JIPM TPM Consistency Award
- **GET Program:** Underwent graduate engineering trainee (GET) program

PUBLICATIONS

- **Sensor identification via Acoustic Physically Unclonable Function (PUF):** Digital Threats: Research and Practice. <https://doi.org/10.1145/3488306>
- **Judicious data management for sustaining an energy harvesting sensor node:** Concurrency and Computation: Practice and Experience. Wiley; doi: 10.1002/cpe.5997

CERTIFICATIONS

- Python Specialization – University of Michigan — 5/5 courses complete - Certificate Link
- Deep Learning – DeepLearning.AI; Ongoing — 3/5 courses complete - Certificates links – 1, 2, 3

VOLUNTEER EXPERIENCE

Yuwa India - Taught middle and high school students at Yuwa School in Hutup Village in Jharkhand. Coached students for tackling interviews. Prepared the team to attend the Laureus Awards 2019, where Yuwa won in "Sports for Good" category.